

CENTRAL BANK OF THE UNITED ARAB EMIRATES

Banking Supervision & Examination Department

SUPERVISORY CIRCULAR

No. 2024/BSE/047

Governance and Risk Management Requirements for Artificial Intelligence and Machine Learning Systems in Licensed Financial Institutions

Circular Reference	CBUAE/BSE/2024/047
Issue Date	15 March 2024
Effective Date	1 July 2024
Addressed To	Chief Executive Officers and Chief Risk Officers of all Licensed Financial Institutions in the UAE
Subject	AI/ML Governance, Model Risk Management, and Consumer Protection Requirements
Supersedes	CBUAE/BSE/2022/031 (Guidance on Automated Decision-Making)
Compliance Deadline	31 December 2024 for Tier 1 institutions; 30 June 2025 for all others

Article 1 — Purpose, Scope and Legal Basis

1.1 This Circular is issued pursuant to the powers vested in the Central Bank of the UAE under Federal Decree-Law No. 14 of 2018 regarding the Central Bank and Organisation of Financial Institutions and Activities, as amended, and the accompanying Executive Regulations.

1.2 This Circular establishes minimum standards for the governance, development, deployment, monitoring, and decommissioning of Artificial Intelligence (AI) and Machine Learning (ML) systems used by Licensed Financial Institutions (LFIs) in the UAE in connection with credit decisions, risk management, pricing of financial products, customer service, fraud detection, and any other material business function.

1.3 This Circular applies to all LFIs licensed and supervised by the Central Bank of the UAE, including but not limited to national banks, foreign bank branches, finance companies, exchange houses, and payment service providers, to the extent that they deploy AI or ML systems in material business processes.

1.4 For the purposes of this Circular, an AI/ML system is considered to be deployed in a 'material business process' where its outputs directly or indirectly influence decisions that: (a) affect a customer's access to or terms of a financial product or service; (b) generate recommendations that are presented to employees or customers as a basis for financial action; or (c) are used to calculate, estimate, or propose prices, rates, fees, or other commercial terms for financial products.

IMPORTANT: The pricing of credit facilities, including the use of AI-generated pricing recommendations for corporate loans, revolving facilities, or any structured credit product, is explicitly classified as a material business process under Article 1.4(c). All requirements in this Circular apply to such systems.

Article 2 — Governance Framework Requirements

2.1 Board and Senior Management Accountability

2.1.1 The Board of Directors of every LFI deploying AI/ML systems in material business processes must adopt and maintain a written AI Governance Policy. This policy must be reviewed and re-approved by the Board at least annually.

2.1.2 A designated member of senior management, at a level no lower than Chief Risk Officer or equivalent, must be appointed as the AI Risk Officer responsible for overseeing the implementation of this Circular.

2.1.3 The AI Governance Policy must address, at minimum: (a) the LFI's risk appetite for AI-related risks; (b) accountability structures for AI development and deployment; (c) human oversight requirements; (d) model inventory management; (e) consumer protection obligations; and (f) regulatory reporting obligations.

2.2 AI Model Inventory

2.2.1 Every LFI must maintain a centralised, up-to-date AI Model Inventory covering all AI/ML systems deployed in material business processes. The inventory must be made available to the Central Bank upon request.

2.2.2 For each model in the inventory, the following minimum information must be recorded:

- Model name, version, and unique identifier.
- Business function and processes in which the model is deployed.
- Date of initial deployment and date of most recent material update.
- Model owner (business function) and model developer (technology/data team).
- Validation status and date of last independent validation.
- Current monitoring status (active / under review / decommissioned).
- Risk classification (High / Medium / Low) as determined under Article 4.

Article 3 — Model Risk Management

3.1 Model Development Standards

3.1.1 All AI/ML models deployed in material business processes must be developed in accordance with sound model development practices, including but not limited to: clearly documented model objectives; appropriate training data governance; out-of-time and out-of-sample testing; and documentation of model limitations and known failure modes.

3.1.2 Training data used for models that influence credit pricing or credit decisions must be sourced from reliable, auditable data systems. Data lineage must be documented from source to model input.

3.2 Independent Model Validation

3.2.1 All High-Risk AI/ML models (as classified under Article 4) must be independently validated before initial deployment and after any material modification. Independent validation means validation conducted by a team or function that is organisationally separate from the model development team.

3.2.2 The Model Validation Report must document: (a) a replication of the model's development methodology; (b) performance benchmarking against alternative approaches; (c) sensitivity analysis; (d) out-of-sample performance testing; and (e) assessment of conceptual soundness of the model's design.

3.2.3 For AI/ML models used in credit pricing, the Model Validation Report must additionally include: (a) a review of the pricing floor and floor compliance mechanism; (b) testing of the human oversight control (see Article 5); and (c) assessment of the model's behaviour under stressed market conditions.

Article 4 — AI Risk Classification

LFIs must classify each AI/ML system in their inventory into one of three risk categories, based on the criteria in Table 4.1.

Risk Class	Criteria	Examples	Validation Required
HIGH	Direct impact on credit decisions, pricing, lending limits, or exclusion/inclusion of customers from financial services.	Credit pricing agents · loan approval models · fraud decisioning systems	Mandatory before deployment; annual re-validation
MEDIUM	Indirect influence on business decisions; used for internal analytics or monitoring with human review of outputs.	Customer segmentation models · churn prediction · internal risk dashboards	Required within 6 months of deployment; biennial
LOW	No direct or indirect decision impact; used for operational efficiency or back-office automation.	Document classification · OCR · internal helpdesk chatbots	Not mandated; recommended at management discretion

CLASSIFICATION GUIDANCE: AI systems used to generate pricing recommendations for corporate credit facilities are HIGH risk under Article 4. This includes large language model (LLM)-based agents that produce pricing proposals, scenario analyses, or rate recommendations, even where human approval is required before the recommendation is acted upon.

Article 5 — Human Oversight and Control Requirements

5.1 Mandatory Human-in-the-Loop Controls

5.1.1 For all HIGH-risk AI/ML systems, LFI must implement mandatory human review and approval controls that ensure no output from the AI/ML system is acted upon without explicit human authorisation by a qualified individual at the appropriate authority level.

5.1.2 The human reviewer must have access to: (a) the AI system's recommendation in full; (b) the underlying data inputs used by the system; (c) the reasoning or explanation provided by the system for its recommendation; and (d) the relevant policy and regulatory context against which the recommendation should be assessed.

5.1.3 LFI must implement technical controls to prevent the bypassing of human review steps. Evidence that the human review was completed, including the identity of the reviewer, the timestamp of review, and the reviewer's decision, must be logged and retained for a minimum of seven years.

5.1.4 Where an AI system is used to generate pricing recommendations for credit facilities, the human reviewer must confirm, in writing or through a system-logged approval, that: (a) the recommended pricing complies with all applicable internal policies; (b) the recommended pricing is not below the regulatory minimum standards; and (c) the reviewer has independently assessed and concurs with the recommendation.

5.2 Override and Escalation

5.2.1 LFI must maintain documented override procedures that allow qualified personnel to deviate from an AI system's recommendation. All overrides must be documented with the reason for override.

5.2.2 Systematic patterns of override (defined as override rates exceeding 25% of recommendations over a rolling 90-day period) must be reported to the AI Risk Officer and reviewed as a potential indicator of model performance degradation.

Article 6 — Explainability and Transparency

6.1 Explainability Requirements

6.1.1 For every recommendation generated by a HIGH-risk AI system, the system must be capable of producing an explanation of the recommendation in plain language. This explanation must be made available to the human reviewer as part of the approval workflow.

6.1.2 For credit pricing AI systems specifically, the explanation must include: (a) the key input signals that drove the recommendation (e.g., FTP rate, RWA, client risk rating, peer benchmarks); (b) a comparison of the recommendation against the applicable pricing floor; (c) an indication of the confidence level of the recommendation; and (d) a summary of any assumptions or data inputs that the system considers uncertain or incomplete.

6.1.3 LFIs must retain records of AI-generated explanations alongside the corresponding recommendations for the same minimum retention period as the underlying credit documentation.

6.2 Customer Transparency

6.2.1 Where the terms offered to a customer (including pricing) have been materially influenced by an AI/ML system, and a customer requests an explanation of how their terms were determined, the LFI must be able to provide a meaningful explanation of the factors that influenced the outcome. The LFI is not required to disclose proprietary model architecture or training data.

Article 7 — Data Governance for AI Systems

7.1 Data Quality

7.1.1 LFIs must implement data quality controls to ensure that data used to train, validate, and operate AI/ML systems is accurate, complete, timely, and fit for purpose. Data quality assessments must be documented and reviewed at least annually.

7.1.2 For AI systems used in credit pricing, the following data governance requirements apply in addition to the general requirements in Article 7.1.1:

- Interest rate and FTP data used as inputs must be sourced directly from authorised internal Treasury systems and must not be manually overridden without a documented audit trail.
- Client risk rating data used as inputs must be sourced from the LFI's credit systems and must reflect the most recently approved internal credit rating.
- Market benchmark data (e.g., peer pricing) used as inputs must be sourced from identified, approved data providers and must be date-stamped.

7.2 Data Retention and Lineage

7.2.1 Complete data lineage — from source system to model input to model output — must be documented and retained for all HIGH-risk AI/ML system predictions. This lineage must support reconstruction of any recommendation for audit or regulatory review purposes.

Article 8 — Regulatory Reporting and Supervisory Access

8.1 LFIs must submit an annual AI/ML Systems Return to the Central Bank of the UAE by 31 March each year, covering the preceding calendar year. The format of the Return will be prescribed by the Central Bank.

8.2 In addition to the annual Return, LFIs must notify the Central Bank within 30 calendar days of: (a) initial deployment of any new HIGH-risk AI/ML system; (b) any material modification to an existing HIGH-risk system; (c) any AI-related incident that results in material consumer detriment or regulatory breach.

8.3 The Central Bank reserves the right to conduct on-site or off-site examinations of LFIs' AI/ML systems, governance frameworks, model documentation, and validation records at any time. LFIs must provide full cooperation with such examinations.

NON-COMPLIANCE: LFIs that fail to comply with the requirements of this Circular by the applicable deadline will be subject to supervisory action, which may include direction to suspend the use of non-compliant AI/ML systems, imposition of administrative penalties, or other measures as provided under Federal Decree-Law No. 14 of 2018.

Article 9 — Transitional Provisions

9.1 LFIs that are currently operating AI/ML systems in material business processes as of the issue date of this Circular must complete a gap assessment against the requirements herein within 60 days of the effective date (1 July 2024) and submit a remediation plan to the Central Bank.

9.2 Existing HIGH-risk systems that have not undergone independent model validation must complete validation by 31 December 2024 for Tier 1 institutions, or by 30 June 2025 for all other LFIs.

Issued under the authority of the Central Bank of the UAE
Banking Supervision & Examination Department | Abu Dhabi, March 2024